Min Woo Park, Ph.D.

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SHORT BIO. During my Ph.D., I focused on *causal decision-making* (NeurIPS-25, UAI-26a, ICLR-26) and *causal effect identification* (NeurIPS-24, UAI-26c). I investigated how to formally model diverse real-world problems within a causal framework and derive optimal decisions from such structured representations.

In particular, I developed methods for robust decision-making under uncertainty (NeurIPS-25), domain shifts (UAI-26a), and even counterfactual actions (ICLR-26).

EDUCATION

Graduate School of Data Science, Seoul National University — Seoul, Republic of Korea

Ph.D. in Data Science — Sep. 2022 – Feb. 2027

Advisor: Prof. Sanghack Lee.
Thesis: *Generalizing Causal Decision-Making: Markov Equivalence, Transportability and Counterfactuals*.
Doctoral Qualifying Exam waived due to outstanding academic performance.

Department of Mathematics and Statistics, Sejong University — Seoul, Republic of Korea

B.S. in Applied statistics and Mathematics (Double major), *Cum Laude* — Mar. 2016 – Aug. 2022

Advisors: Prof. Ji Eun Lee and Jin Hee Yoon. — (2.5 years military service)

RESEARCH EXPERIENCE

Samsung Research | AI Research Engineer — Starting in Mar. 2027

Furiosa AI | AI Research Engineer Internship — Jun. 2026 – Sep. 2026

Causality Lab, Seoul National University | Ph.D. Researcher — Jan. 2023 – Feb. 2027

Functional Analysis Lab, Sejong University | Undergraduate Researcher — Jul. 2021 – Dec. 2022

ISDS Lab, Sejong University | Undergraduate Researcher — Jan. 2021 – Dec. 2022

PUBLICATIONS

* for joint authorship and † for corresponding author.

◇ On Transportability for Structural Causal Bandits.
Min Woo Park and Sanghack Lee†. *Uncertainty in Artificial Intelligence* (UAI), 2026.

◇ Breaking Bad: Component-wise Parent Deletion for Score-Based Causal Discovery.
Min Woo Park*, Taehui Yun*, Youngin Jang, Younseok Yeom, Jonghwan Kim, Jiyeon Kang, Songseong Kim, Hyemin Jung, Sangmin Lee, Jongseong Jang†, and Sanghack Lee†. *Uncertainty in Artificial Intelligence* (UAI), 2026.

◇ Canonical Domain Reduction for Partial Counterfactual Identification.
Yesong Choe, Yeahoon Kwon*, **Min Woo Park***, and Sanghack Lee†. *Uncertainty in Artificial Intelligence* (UAI), 2026.

◇ Counterfactual Structural Causal Bandits.
Min Woo Park and Sanghack Lee†. *International Conference on Learning Representations* (ICLR), 2026.

◇ Structural Causal Bandits under Markov Equivalence.
Min Woo Park, Andy Arditi, Elias Bareinboim†, Sanghack Lee†. *Neural Information Processing Systems* (NeurIPS), 2025
CausalAI Laboratory, Columbia University, Technical Report (R-122).

◇ Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference.
Yonghan Jung†, **Min Woo Park**, and Sanghack Lee†. *Neural Information Processing Systems* (NeurIPS), 2024.

◇ Computation of the iterated Aluthge, Duggal, and Mean transforms.
Min Woo Park*, and Ji Eun Lee*†. *Filomat*, 2023.

◇ Convergence of the iterated mean transforms of a 2×2 matrix
Ji Eun Lee*† and **Min Woo Park***. under review.

AWARDS AND HONORS

NRF Ph.D. fellowship in science and engineering, 2026 | National Research Foundation of Korea

- ▹ *Active Learning-Based Action Space Optimization for Causal Multi-Armed Bandits under Incomplete Information*. Sep.2026 – Aug.2027

UAI 2026 scholarship Aug. 2026

Graduate Student Instructor (GSI) scholarship Spring 2024, Spring 2025, Spring 2026

BK21 FOUR scholarship Fall 2024, Fall 2025

First place merit scholarship | Sejong University Fall 2020, Fall 2021

Fourth place award | The 9th College of Natural Sciences Academic Conference, Sejong University

- ▹ *Mitigating Spurious Regression in Time-Series Data via Granger Causality*. Nov. 2021

Third place award | The 8th College of Natural Sciences Academic Conference, Sejong University

- ▹ *The Significance of the Euler's Number and Its Applications*. Nov. 2020

PROJECTS

Towards More Scalable Score-based Causal Discovery | LG AI Research Sep. 2025 – Aug. 2026

- ▹ Leading a project on scalable score-based causal discovery algorithms.
  - Developed a novel operator enabling score-based causal discovery algorithms to escape local optima.
  - Investigating parallelization strategies for score-based causal discovery algorithms.
- ▹ Resulting paper accepted at *Uncertainty in Artificial Intelligence* (UAI-26b).
  - Software available at `https://github.com/LGAI-Research/breaking-bad`.
- ▹ Korean patent application filed.

Hyundai NGV Data Boot-Campus | Hyundai Motor Company

- ▹ **2025.** Developed high-voltage battery performance prediction technology. Jul. 2025 – Sep. 2025
- ▹ **2024.** Developed an intelligent system for predicting vulnerable regions of automotive body panel stiffness using differential geometrybased automatic curvature analysis. Jul. 2024 – Sep. 2024

Optimal Mixed Policies Completeness | Oxford University Aug. 2023 – Feb. 2024

- ▹ Studied characterization of conditions under which an intervention effect of a variable is positive.
- ▹ Acknowledged for helpful comments in *Toward a Complete Criterion for Value of Information in Insoluble Decision Problems*. *Transactions on Machine Learning Research* (TMLR), 2024.

TEACHING ASSISTANTS

Data Science Project Fall 2026

- ▹ Practical experience on end-to-end project execution.

Causal Inference for Data Science Spring 2024, Fall 2024, Spring 2025, Spring 2026

- ▹ Topics included Bayesian networks and graphical causal inference.

Machine Learning and Deep Learning for Data Science II Fall 2023

- ▹ Covered kernel methods, Gaussian processes, and related topics in machine learning.

Basic Mathematics for Artificial Intelligence (Undergraduate) Spring 2022

- ▹ Covered linear algebra, basic statistics, and probability theory.

ACADEMIC SERVICES, PROFESSIONAL TALKS AND POSTERS

Academic Services

Conference Reviewer: **2027** ICLR, **2026** NeurIPS, UAI, ICLR, ICML, **2025** ICML

Reviewed AI/ML lecture video content from LG AI Research for LG Aimers
Summer 2025, Winter 2025, Summer 2026

Invited Talks

Major Exploration for Sejong University Students | Seoul, Republic of Korea	Nov. 2026 (scheduled)
SNU GSDS student research seminar | Seoul, Republic of Korea	Dec. 2024, Apr. 2025

Poster Sessions

UAI 2026 poster session | Amsterdam, Netherlands	Aug. 2026
ICLR 2026 poster session | Rio de Janeiro, Brazil	Apr. 2026
ExploreCSR poster session | Seoul, Republic of Korea	Dec. 2025
NeurIPS 2025 poster session | San Diego, USA	Dec. 2025
JKAIA 2025, NeurIPS preview session | Seoul, Republic of Korea	Nov. 2025
SNU GSDS workshop poster session | Jeju, Republic of Korea	Jul. 2025
NeurIPS 2024 poster session | Vancouver, Canada	Dec. 2024

MENTORING

Taehui Yun | Graduate School of Data Science, Seoul National University — dbs4835@snu.ac.kr
Ph.D. student. Co-advised a project; paper accepted at UAI 2026 (UAI-26b).

Youngin Jang | Graduate School of Data Science, Seoul National University — youngijang@snu.ac.kr
M.S. student. Co-advised a project; paper accepted at UAI 2026 (UAI-26b).

REFERENCE

Sanghack Lee | Graduate School of Data Science, Seoul National University — sanghack@snu.ac.kr
Associate professor

Ji Eun Lee | Department of Mathematics and Statistics, Sejong University — jieunlee7@sejong.ac.kr
Associate professor

Jin Hee Yoon | Dept. of AI and Information Technology, Sejong University — jin9135@sejong.ac.kr
Associate professor

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